A Riesz representer perspective on targeted learning

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Causal Inference
Modus Operandi

1. Choose an estimand ψ and gather data $O_1, \ldots, O_n \sim \mathsf{P}$

2. Construct an estimator of ψ

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Targeted Minimum Loss-Based Estimation

Even if P totally unknown (nonparametric), can construct "good" plug-in estimator of ψ by...

1. Deriving an "efficient influence function" $\phi(\mathsf{P})(O_i; \psi)$

2. **TMLE**: Choosing and optimizing loss $L(O, \varepsilon)$ satisfying, for some v,

$$v^\top \nabla_\varepsilon L(O; \varepsilon)\Big|_{\varepsilon=0} = \frac{1}{n}\sum_{i=1}^{n} \phi(\mathsf{P}_n)(O_i; \psi)$$

e.g. choose L to respect problem constraints (i.e. bounded outcomes)

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TMLE.jl

Search docs (Ctrl + /)

R/txshift

Efficient Estimation of the Causal

R/tmle3mediate

Targeted Learning for Causal Mediation Analysis

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Itmle: An R Package Implementing Targeted Minimum Loss-Based Estimation for Longitudinal Data

Samuel D. Lendle, Joshua Schwab, Maya L. Petersen, Mark J. van der Laan

Authors: Nima Hejazi, James Duncan, David McCoy, and Mark van der Laan

Q: Can we derive a TMLE for a *general* class of estimands?

Estimand Name	**Formula for ψ**
Counterfactual mean (Static)	$\mathrm{E}[\mathrm{E}(Y \mid A = a, L)]$
Counterfactual mean (Stochastic)	$\mathrm{E}[\mathrm{E}(Y \mid A = A^{\star}, L)]$
Average quantile effect	$\mathrm{E}[Q_{\tau}(Y \mid A = 1, L) - Q_{\tau}(Y \mid A = 0, L)]$
Average treatment effect on treated	$\mathrm{E}(\mathrm{E}(Y \mid A = 1, L) \mid A = 1)$
Mediational effect	$\mathrm{E}[\mathrm{E}(\mathrm{E}(Y \mid A = 1, M, L) \mid A = 0, L)]$
Longitudinal treatment-specific mean	$\mathrm{E}[\mathrm{E}(\ldots \mathrm{E}(Y \mid \bar{A}_t = \bar{a}_T, \bar{L}_T) \ldots \mid A_1 = a_1, L_1, L_0)]$
Longitudinal modified treatment effect	$\mathrm{E}[\mathrm{E}(\ldots \mathrm{E}(Y \mid \bar{A}_t = \bar{A}_t^d, \bar{L}_T) \ldots \mid A_1 = A_1^d, L_1, L_0)]$

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Estimand Name	**Formula for ψ**
Counterfactual mean (Static) Counterfactual mean (Stochastic) Average quantile effect Average treatment effect on treated Mediational effect Longitudinal treatment-specific mean Longitudinal modified treatment effect	$\mathrm{E}[h(O;\eta)]$

data

$$\mathbb{E}[h(O;\eta)]$$

function "evaluator"

nuisance function

Riesz Representation

Riesz Representation Theorem (statistics version).

Suppose $\eta \in \mathcal{L}_2(\mathsf{P})$, and $\psi := \Psi(\eta) = \mathsf{E}(h(O;\eta))$ is a bounded linear functional. Then, there exists a **Riesz representer** $\alpha \in \mathcal{L}_2(\mathsf{P})$ such that

$$\Psi(\eta) = \mathsf{E}[\alpha(O)\eta(O)]$$

Think of α like a "balancing weight"!

Theorem (Riesz EIF). The efficient influence function of $\Psi(\eta)$ is

$$\phi(\mathsf{P})(O) = \underbrace{h(O;\eta) - \Psi(\eta)}_{\text{expected value EIF}} + \underbrace{\int \alpha(O)\phi_\eta(\mathsf{P})(O)d\mathsf{P}}_{\text{"reweighted nuisance bias"}}$$

where $\phi_\eta(\mathsf{P})(O)$ denotes the efficient influence function of the nuisance parameter η.

*Generalizes previous work, like Hirshberg and Wager (**2021**), Chernozhukov et al. (**2022**), or Williams et al. (**2025**)*

Corollary (Riesz EIF with conditional mean nuisance). Suppose the observed data is $O = (L, A, Y)$, and $\eta(A, L) = E(Y \mid A, L)$.

Then, the efficient influence function of $\Psi(\eta)$ is

$$\phi(\mathsf{P})(O) = h(A, L; \eta) - \Psi(\eta) + \alpha(A, L)(Y - \eta(A, L))$$

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Proof. Just plug in to previous theorem the known efficient influence function of conditional mean

$$\phi_\eta(O) = \frac{\delta_{A,L}}{d\mathsf{P}(A, L)}(Y - E(Y \mid A, L))$$

Side note: Also works with other nuisances. Consider instead treatment-specific τ-quantile effect $\Psi(\eta) = E[Q^\tau(a, L)]$, where $\eta(O) = Q^\tau(A, L)$. By the same logic, the EIF of $\Psi(\eta)$ is

$$\phi(P)(O) = h(O; Q^\tau) - \Psi(\eta) + \alpha(A, L) \underbrace{\left(\frac{\tau - \mathbb{1}(Y > Q^\tau(A, L))}{d\mathsf{P}(Q^\tau(A, L) \mid A, L)} \right)}_{\text{reweighted "pinball loss" derivative}}$$

Many more ways to use this

Consider a time-ordered data structure

$$O = (L_1, A_1, \dots, L_T, A_T, Y)$$

Denote the histories at time t as $\bar{A}_t$ and $\bar{L}_t$. Encompasses:

- Longitudinal data
- Mediation
- Multi-phase sampling

Theorem (Sequential Riesz EIF)

Consider the estimand $\Psi(\eta_1) = \mathsf{E}_{\mathsf{P}}[h_1(\bar{A}_1, \bar{L}_1; \eta_1)]$, where η_t is a bounded linear functional defined sequentially such that, for $t = 1, \ldots, T$, we have

$$\eta_t(\bar{A}_t, \bar{L}_t) = \mathsf{E}[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \eta_{t+1}) \mid \bar{A}_t, \bar{L}_t]$$

with $h_T(\bar{A}_{T+1}, \bar{L}_{T+1}; \eta_{T+1}) := Y$. Let α_t denote the Riesz representer for η_{t+1} in the functional $\mathsf{E}[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \eta_{t+1}) \mid \bar{A}_t, \bar{L}_t]$. Then, the EIF the estimand $\Psi(\mathsf{P}_0)$ is

$$\underbrace{h_1(\bar{A}_1, \bar{L}_1; \eta_1) - \Psi(\eta)}_{\text{Expected value EIF}} + \sum_{t=1}^{T} \underbrace{\prod_{k=1}^{t} \alpha_t(\bar{A}_t, \bar{L}_t)}_{\text{reweighting}} \underbrace{[h_{t+1}(\bar{A}_{t+1}, \bar{L}_{t+1}; \eta_{t+1}) - \eta_t(\bar{A}_t, \bar{L}_t)]}_{\text{Residuals of sequential regressions}}$$

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Sequential TMLE

1. Fit sequential regressions $\eta_1, \dots, \eta_T$ and Riesz representers $\alpha_1, \dots, \alpha_T$.

2. For $t = 1, \dots T$, compute the weights $\omega_t(\bar{A}_t, \bar{L}_t) = \prod_{k=1}^{t} \alpha_k(\bar{A}_k, \bar{L}_k)$

3. For $t = T-1, \dots, 1$, fit 1-D parametric model $\eta_{t,\hat{\varepsilon}_t}$ that regresses
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and set $\eta_t = \eta_{t,\hat{\varepsilon}_t}$

4. The final TMLE update is the updated plug-in estimator

$$\psi_n = \frac{1}{n} \sum_{i=1}^{n} \text{link}[h_1(\bar{A}_{i1}, \bar{L}_{i1}; \eta_{1,\hat{\varepsilon}_1})]$$

→ *Consistent, asymptotically normal, and semi-parametric efficient*

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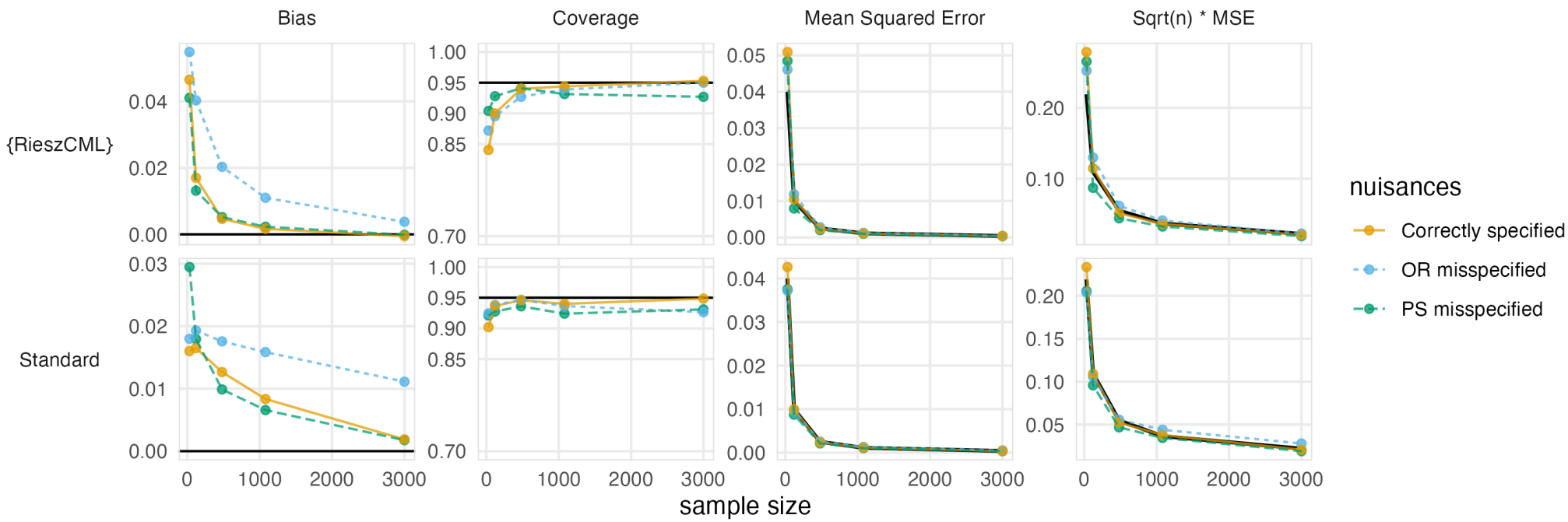
The `RieszCML` package: Simulations

Convergence diagnostics for {RieszCML} TMLE and standard software

Black lines indicate ideal bias, coverage, and the efficiency bound.



All experiments were run for 2,000 replications.

Conclusions

- Unifies semi-parametric efficient estimation across…
  - → *Data*: Longitudinal, mediation, multi-phase sampling, etc.
  - → *Interventions*: Binary, stochastic, derivatives, quantile effects, etc.
- Support theory and software re-use (*Lego bricks*)
- Agnostic about how α is learned; see, e.g. *Riesz regression* (Chernozhukov *et al.*, 2022)

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Thank you! Questions?

Preprint

More about me

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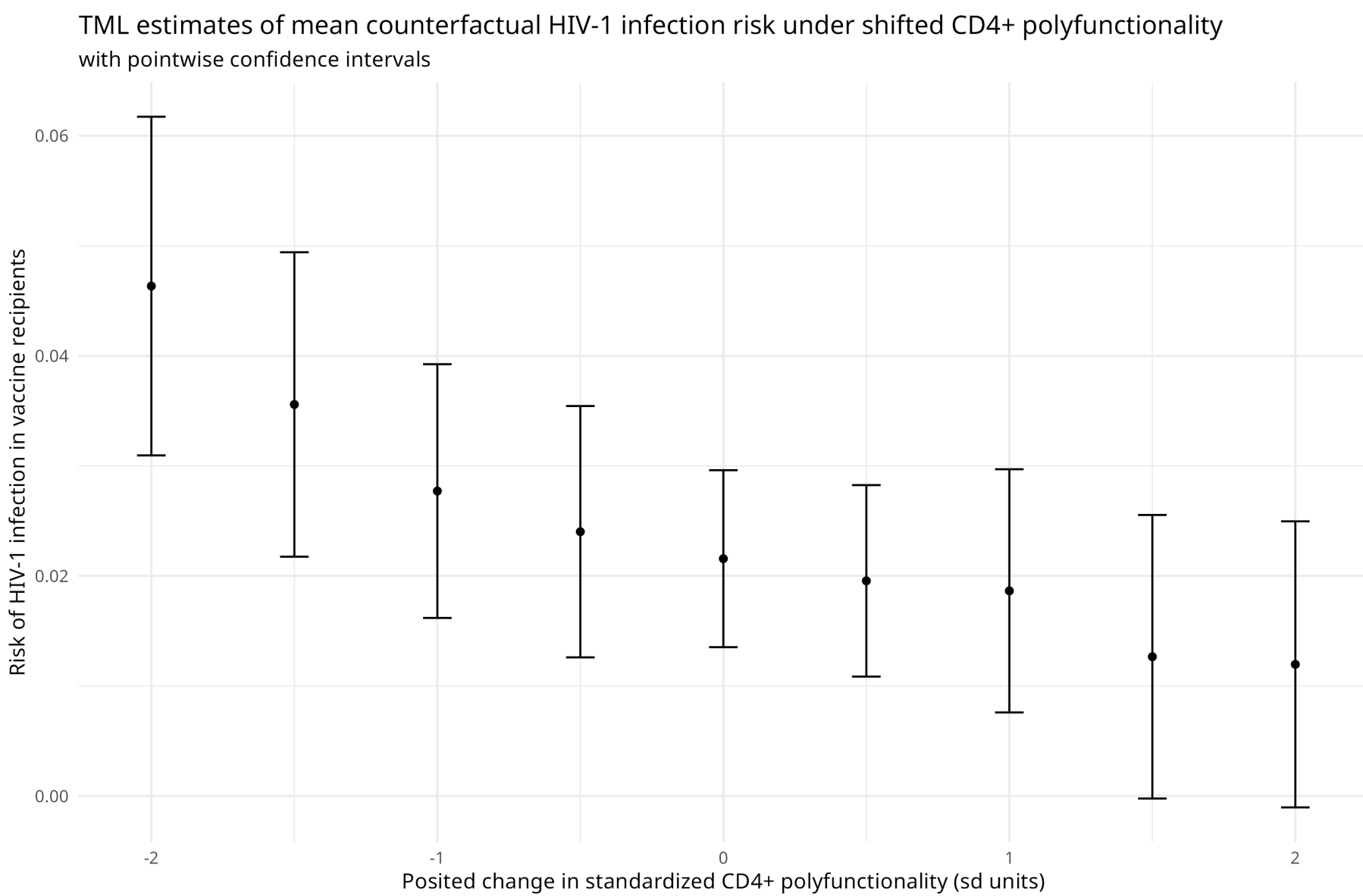
Appendix A: Data Analysis



Appendix B: Functional analysis

Theorem (Riesz Representation, general).

Suppose $\eta \in \mathcal{H}$, a Hilbert space, and that $\psi(\eta) : \mathcal{H} \mapsto \mathbb{R}$ is a bounded linear functional.

Then, there exists $\alpha \in \mathcal{H}$ such that

$$\psi(\eta) = \langle \alpha, \eta \rangle$$